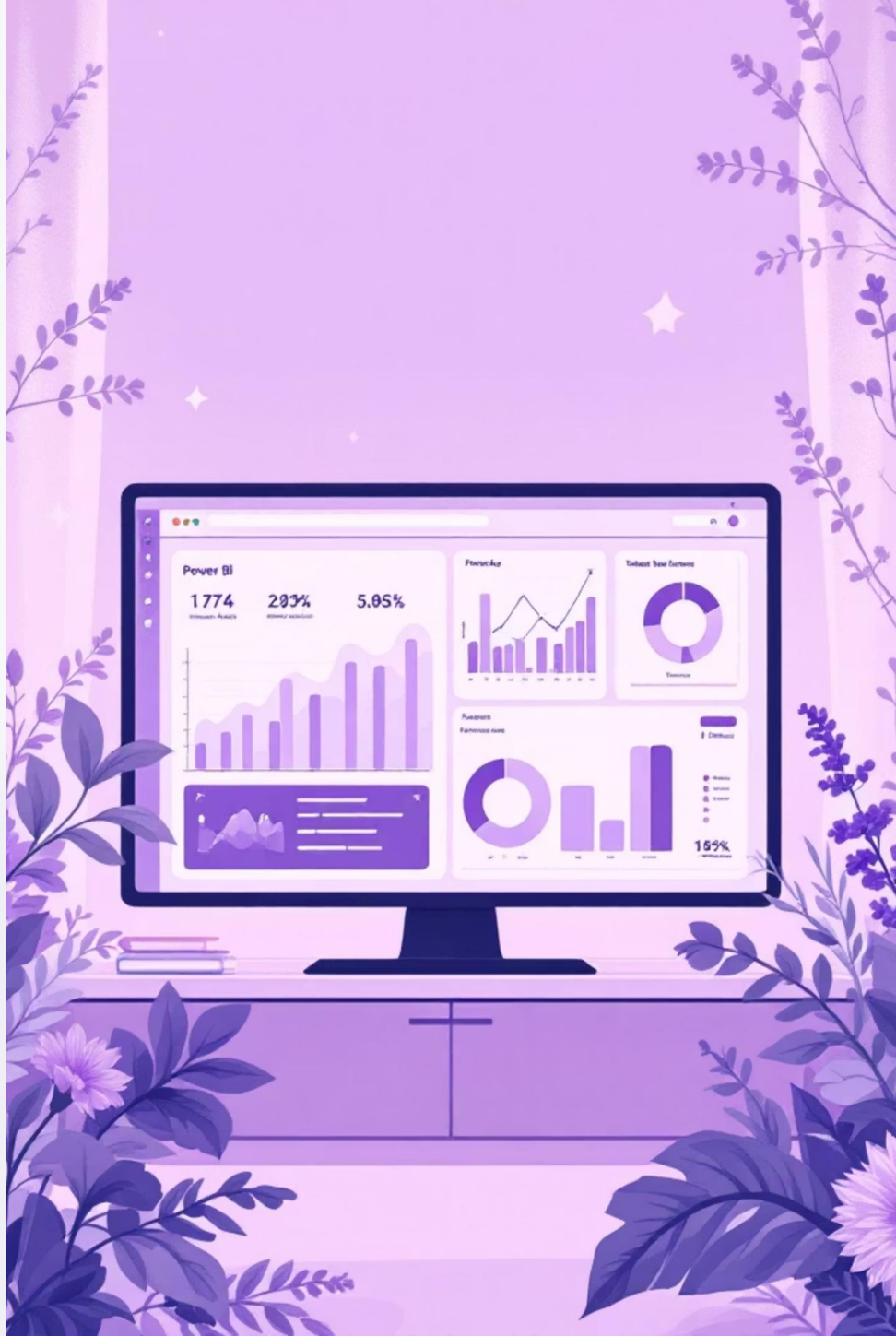
Adventure Works Sales & Business Performance Analysis

Tools Used: Excel, Power BI, Tableau, & SQL

Group 6 :

- Aayush Mevati
- Akshay Bangar
- Karthik S
- Mohammad Abdur Rahman
- Sathish P



Project Objectives

- Analyze overall business performance
- Identify sales and profit drivers
- Understand customer behavior patterns
- Evaluate regional performance
- Support strategic business decision-making using dashboards

Business KPI Overview

29.36M	12.08M	41%	60K
Total Sales	Total Profit	Profit Margin	Total Orders

18.48K	606	10
Customers	Products	Regions

Insight: The company operates with strong profitability and efficient cost management.

Sales Growth Analysis

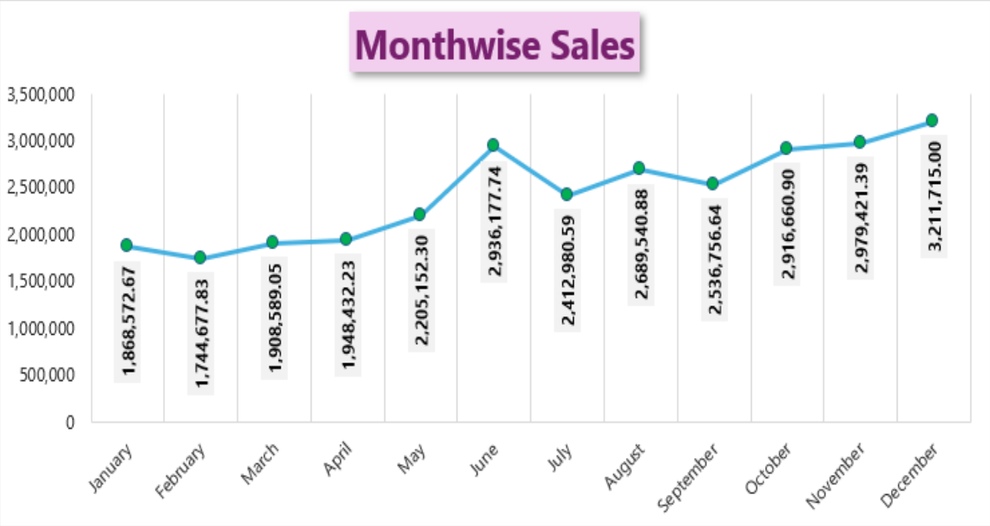
- Significant growth observed from 2011 to 2013
- Peak sales achieved in 2013
- Early years represent expansion phase

Insight: Business experienced rapid scaling and successful market adoption.

Seasonal Sales Trends

- Sales increase progressively through the year
- Highest performance in November and December
- Q4 contributes the largest revenue share (~31%)
- February records lowest sales

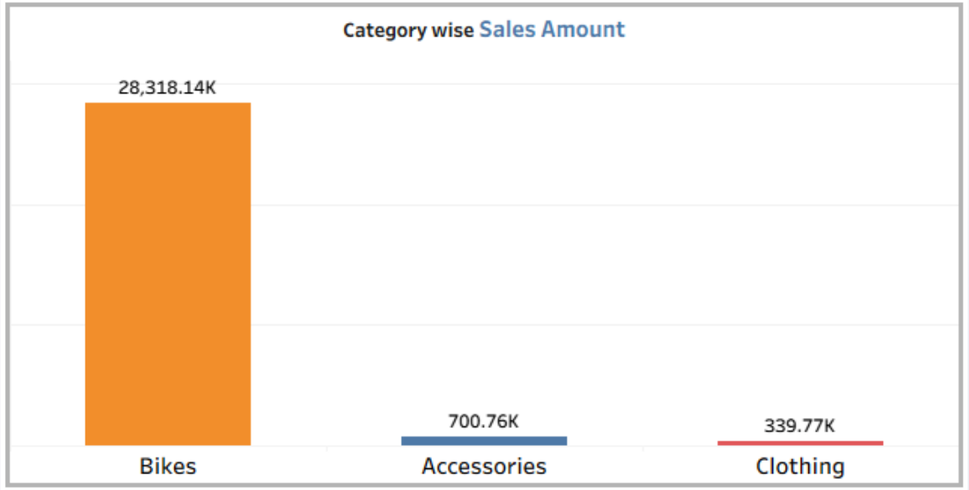
Insight: Sales demonstrate strong seasonal demand patterns.

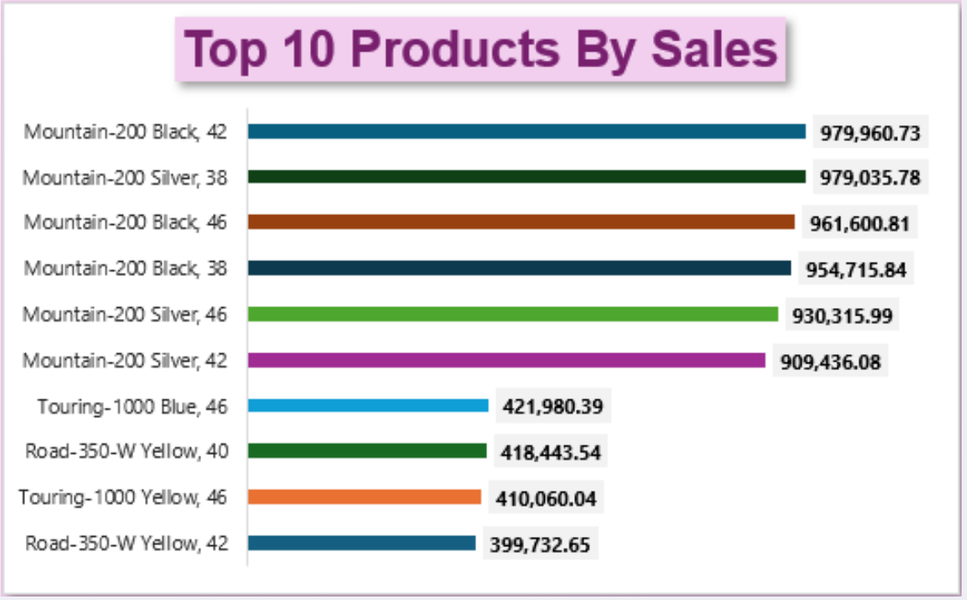


Product Contribution Analysis

- Bikes category dominates revenue contribution
- Accessories and clothing contribute smaller shares
- Top-performing products include Mountain-200 and Road bike models

Insight: Premium bike products are the primary revenue drivers.





Regional Sales Performance

Top Markets

United States: 9.39M

Australia: 9.06M

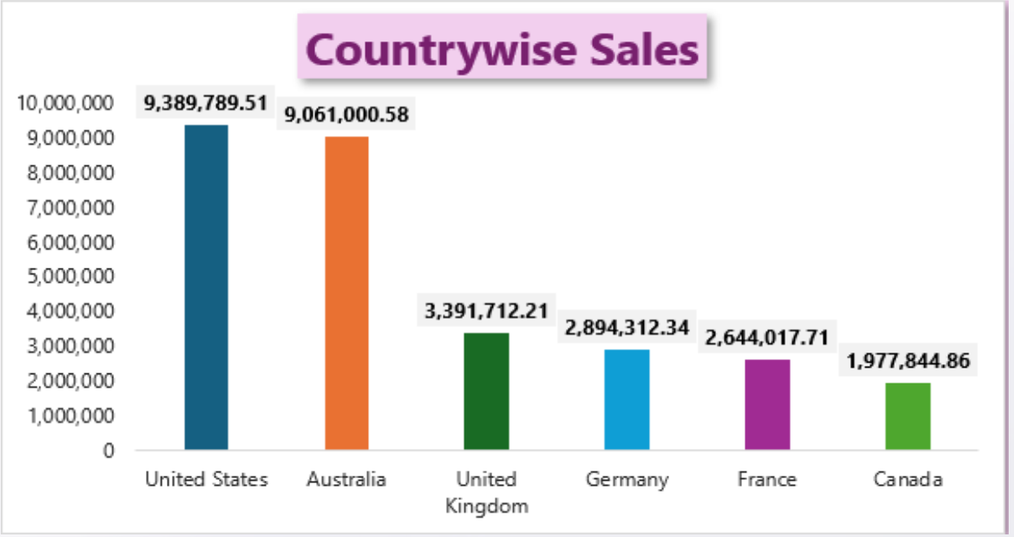
Lower Performing Markets

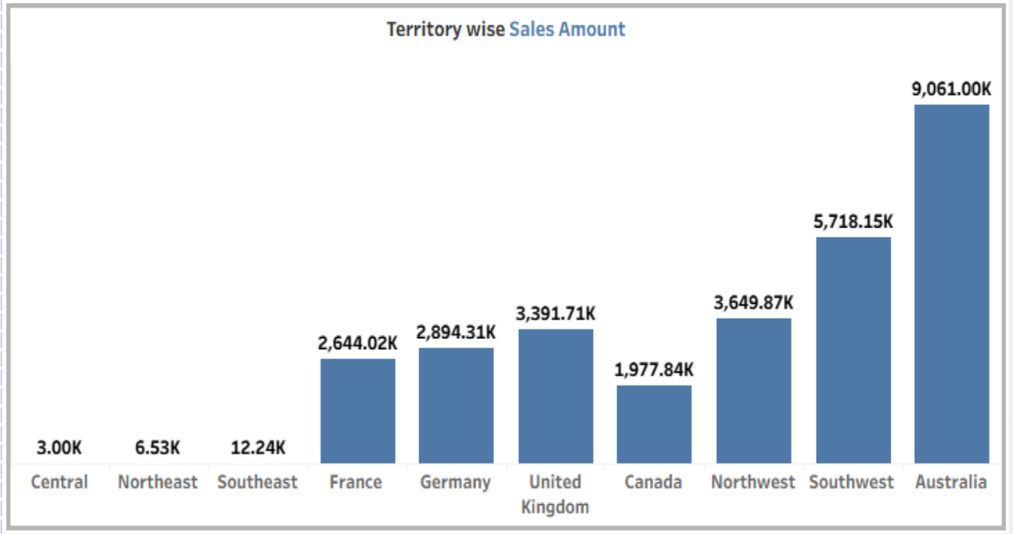
Canada and smaller territories

Market Opportunity

Expansion opportunities elsewhere

Insight: Revenue is concentrated in North America and Australia, creating expansion opportunities elsewhere.

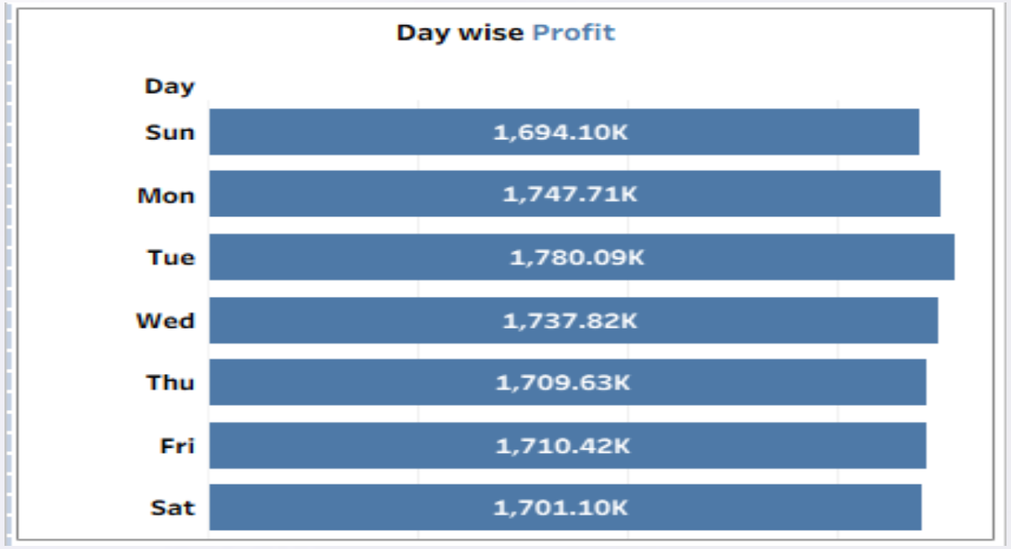




Profitability Trends

- Profit closely follows sales growth
- Highest profit recorded in 2013
- Weekdays profits higher than weekends

Insight: Operational efficiency maintained during business expansion.



Customer Behavior Analysis

Gender Distribution

Nearly equal gender distribution among customers

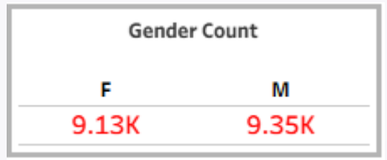
Sales Contribution

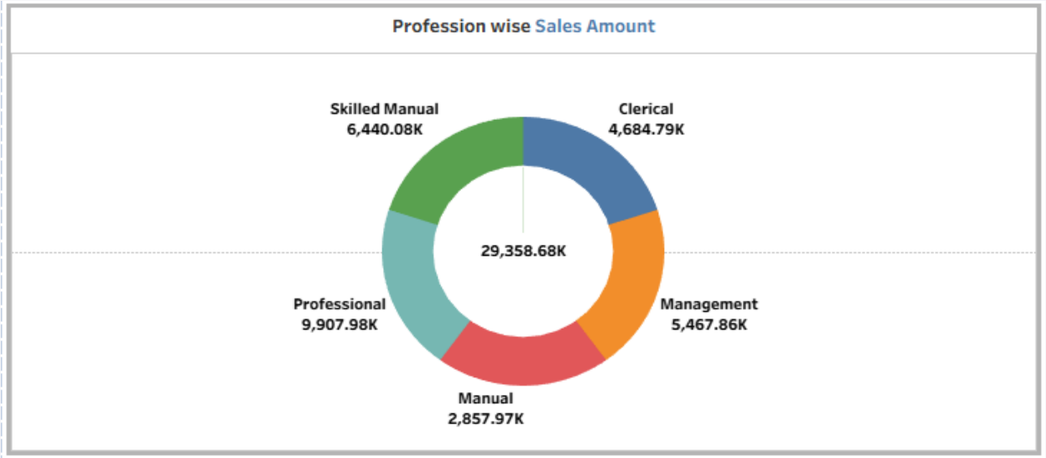
Professionals generate highest sales contribution

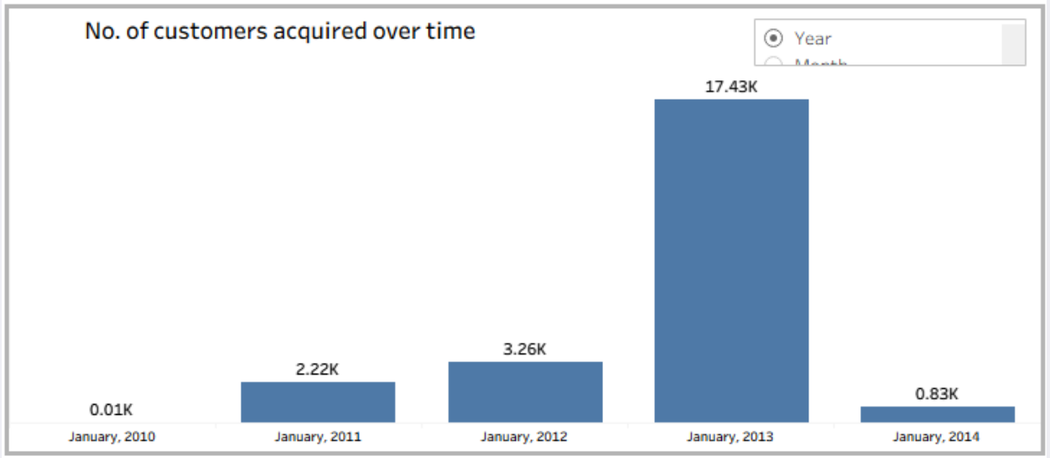
Acquisition Peak

Customer acquisition peaked during 2013

Insight: Target market consists primarily of middle and high-income professionals.







Key Findings & Recommendations

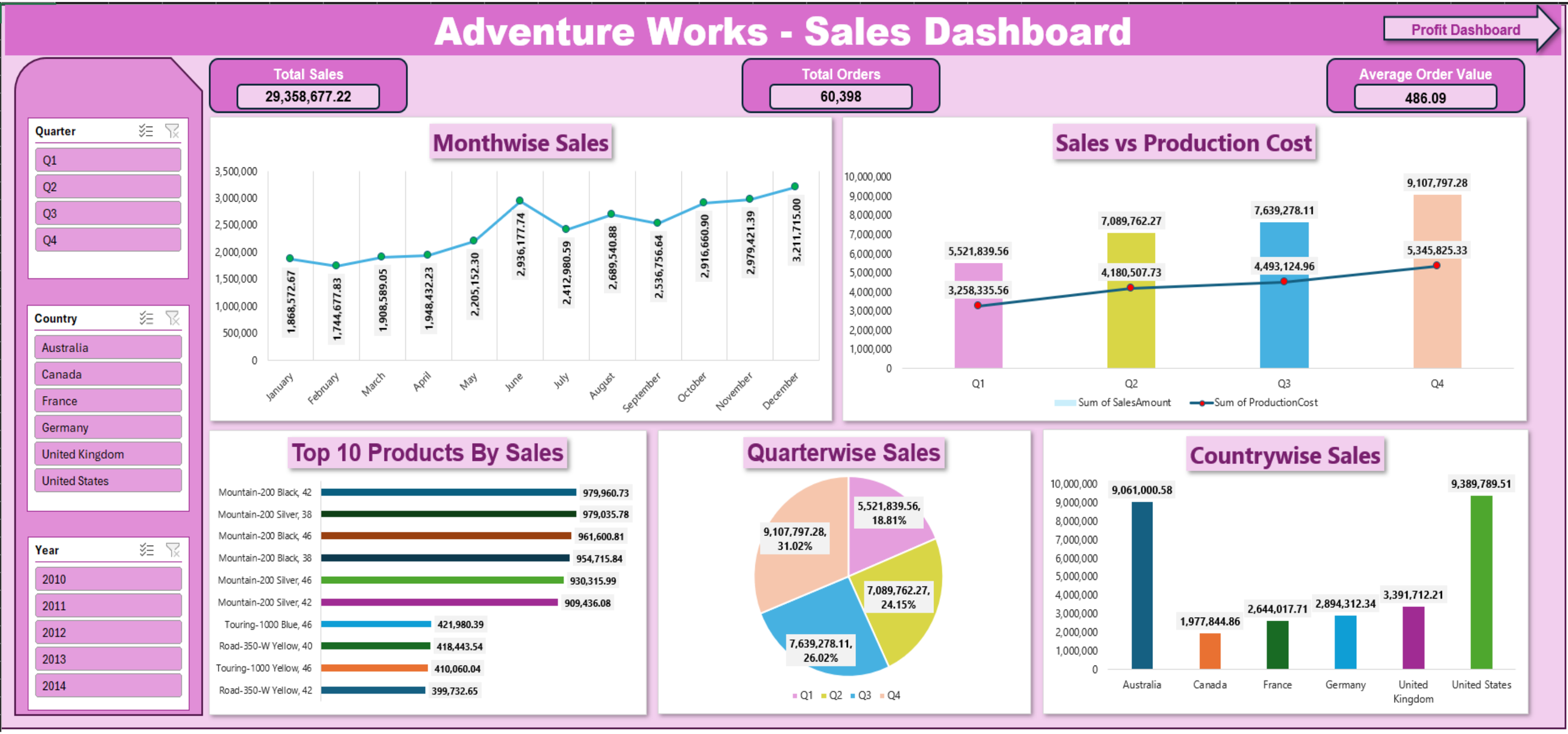
Key Findings:

- High profitability business model
- Premium products drive revenue
- Strong seasonal demand
- Professionals are core customer segment
- Geographic revenue concentration exists

Recommendations:

- Expand European markets
- Improve accessory cross-selling strategy
- Focus marketing during Q4 peak season
- Maintain premium product positioning

Excel Dashboards



Sales Dashboard

Adventure Works - Profit Dashboard

Production Cost
17,277,793.58

Total Profit
12,080,883.65

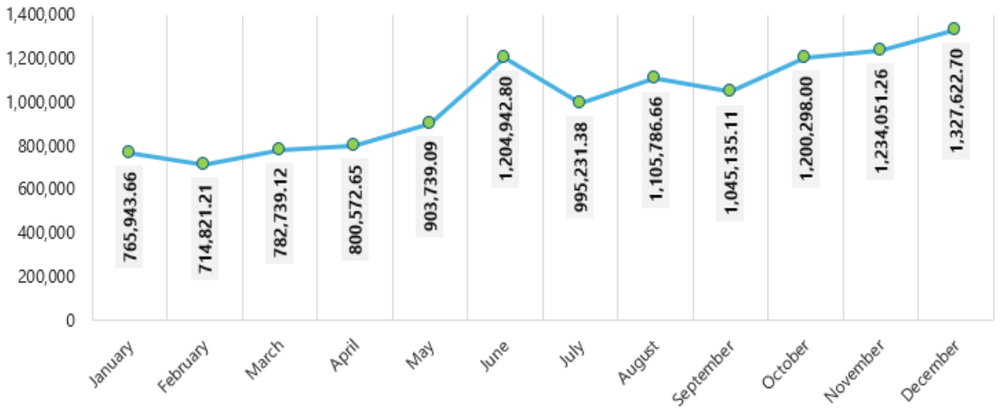
Profit Margin %
41.15%

Quarter
- Q1
- Q2
- Q3
- Q4

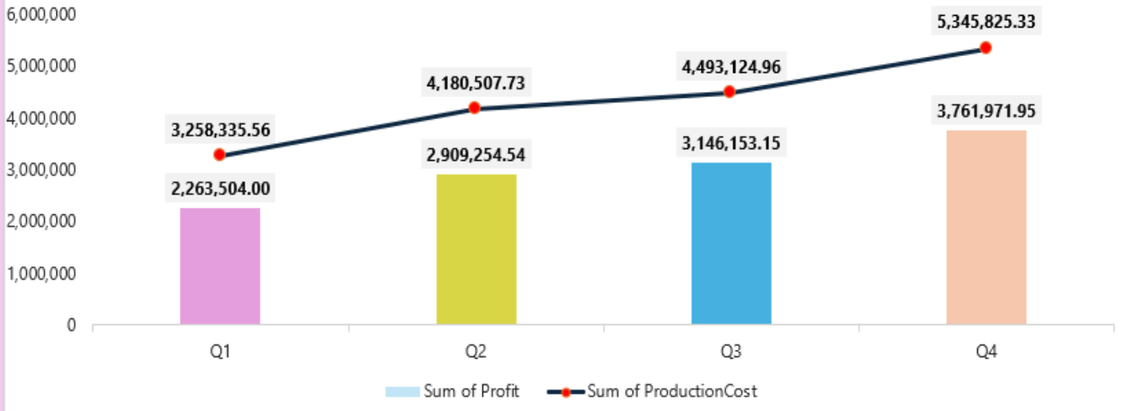
Country
- Australia
- Canada
- France
- Germany
- United Kingdom
- United States

Year
- 2010
- 2011
- 2012
- 2013
- 2014

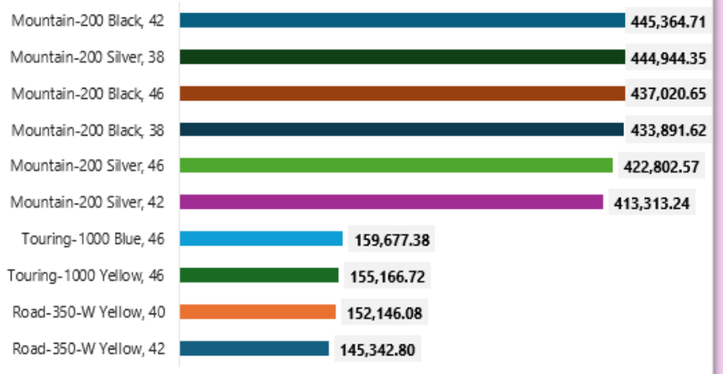
Monthwise Profit



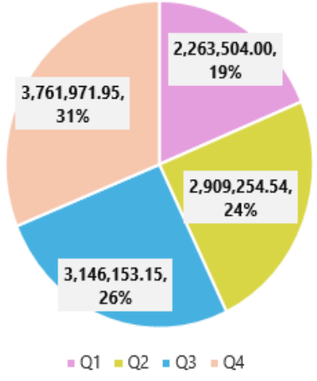
Profit vs Production Cost



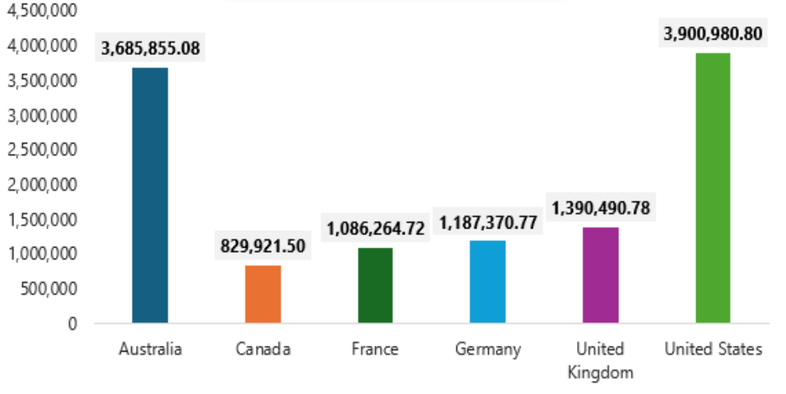
Top 10 Products By Profit



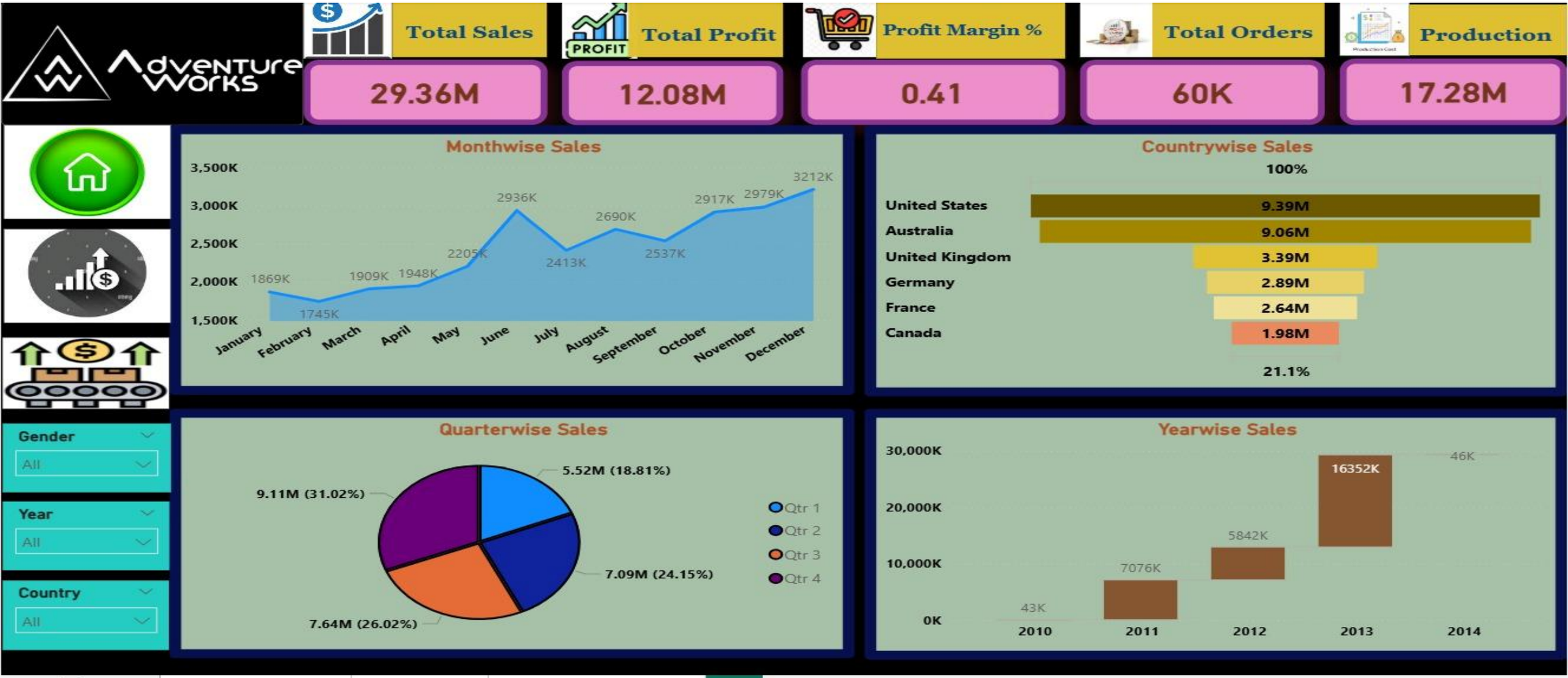
Quarterwise Profit



Countrywise Profit



Power BI Dashboards



Home-Saleswise | Profitwise | Profit by Production

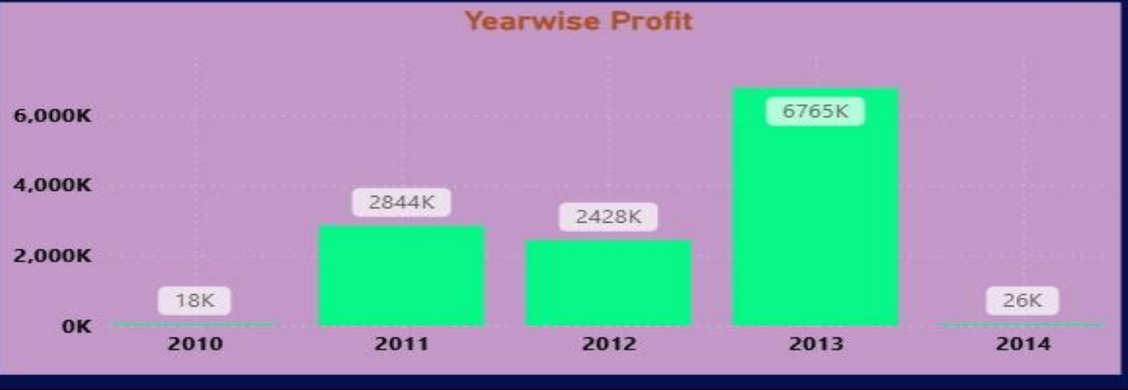
Adventure Works

Total Sales	Total Profit	Profit Margin %	Total Orders	Production
29.36M	12.08M	0.41	60K	17.28M

Gender
All

Year
All

Country
All



Home-Saleswise | Profitwise | Profit by Production

Adventure Works

Total Sales	Total Profit	Profit Margin %	Total Orders	Production
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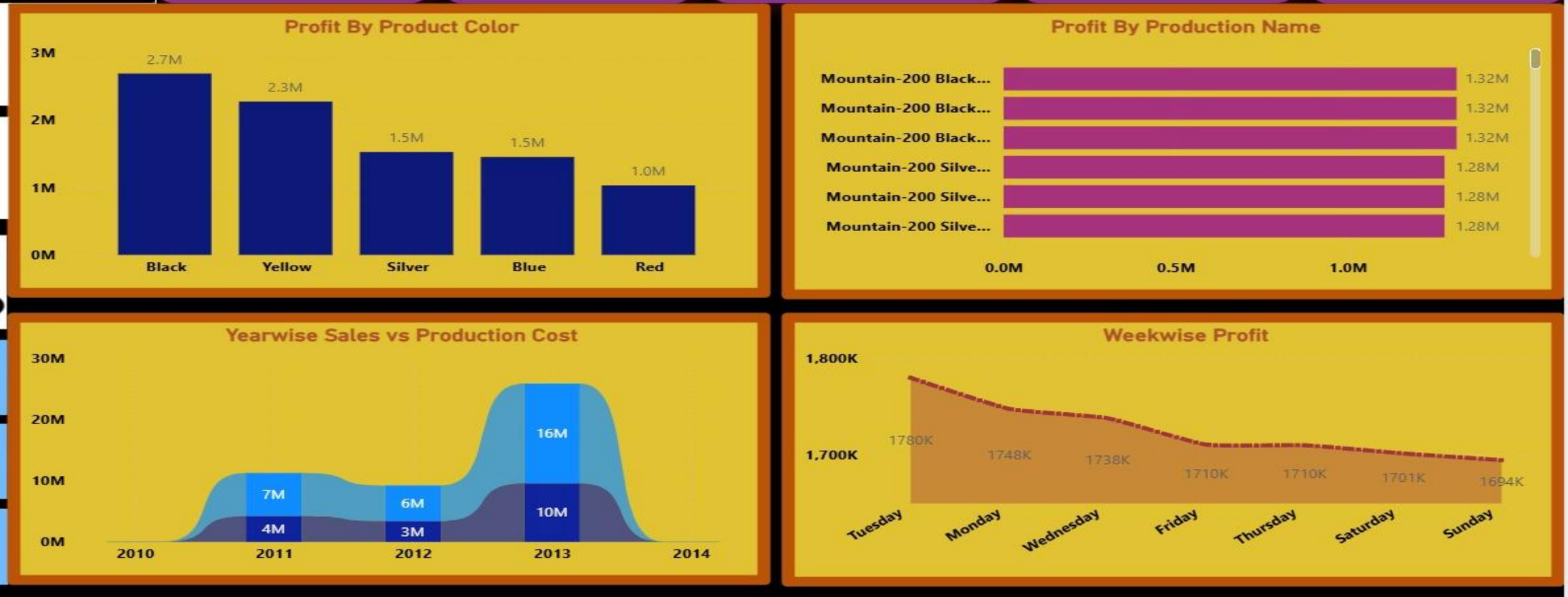

Gender: All

Year: All

Country: All

Home-Saleswise | Profitwise | Profit by Production

Tableau Dashboards

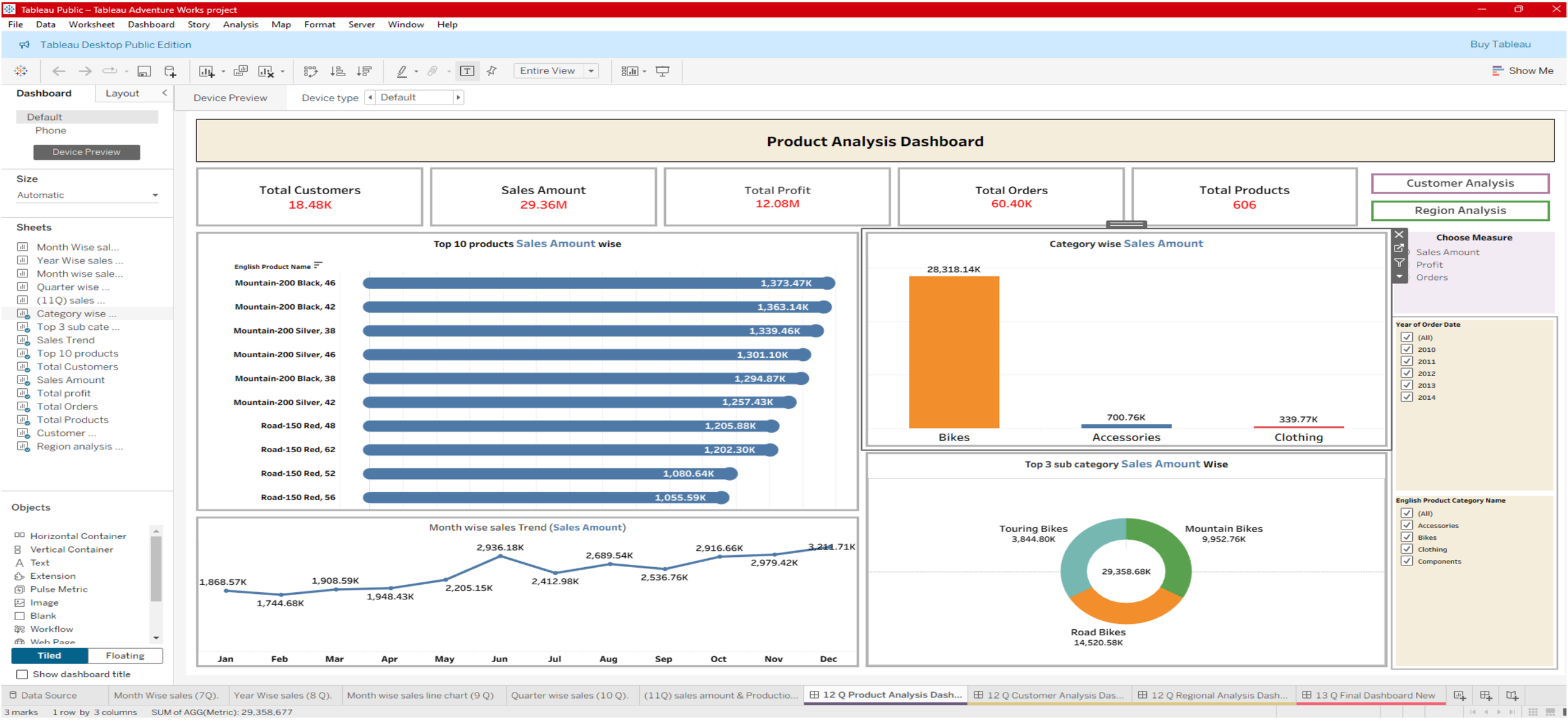

Tableau Public – Tableau Adventure Works project

File Data Worksheet Dashboard Story Analysis Map Format Server Window Help

Tableau Desktop Public Edition | Buy Tableau

Show Me

Dashboard | Layout

Default
Phone
Device Preview

Size
Automatic

Sheets
- Month Wise sal...
- Year Wise sales ...
- Month wise sale...
- Quarter wise ...
- (11Q) sales ...
- Total Customers
- Sales Amount
- Total profit
- Total Orders
- Total Products
- Profession wise ...
- time /cust ...
- Top 15 ...
- product analysi...
- Region analysis ...
- Gender c
- Gender Count

Objects
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- Vertical Container
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- Extension
- Pulse Metric
- Image
- Blank
- Workflow
- Web Page

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Show dashboard title

Device Preview | Device type: Default

Customer Analysis Dashboard

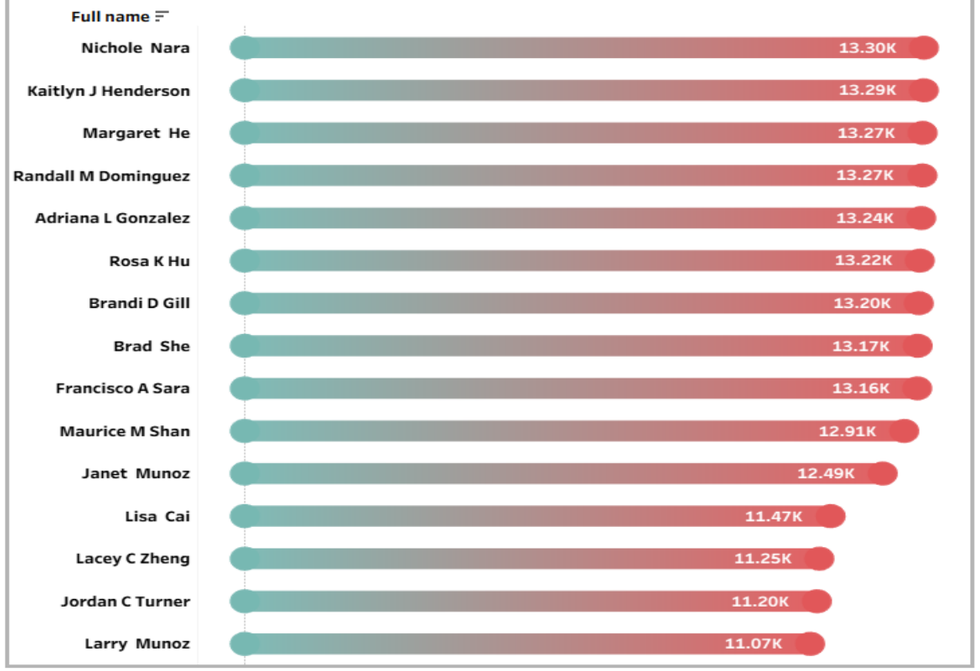
Gender Count	
F	M
9.13K	9.35K

Total Customers	Sales Amount	Total Profit	Total Orders	Total Products
18.48K	29.36M	12.08M	60.40K	606

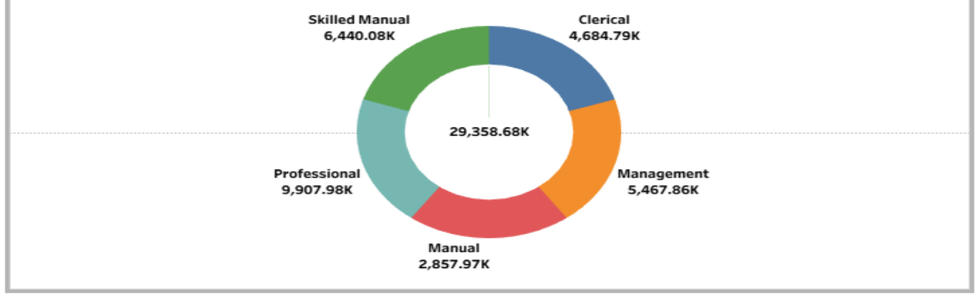
Product Analysis

Region Analysis

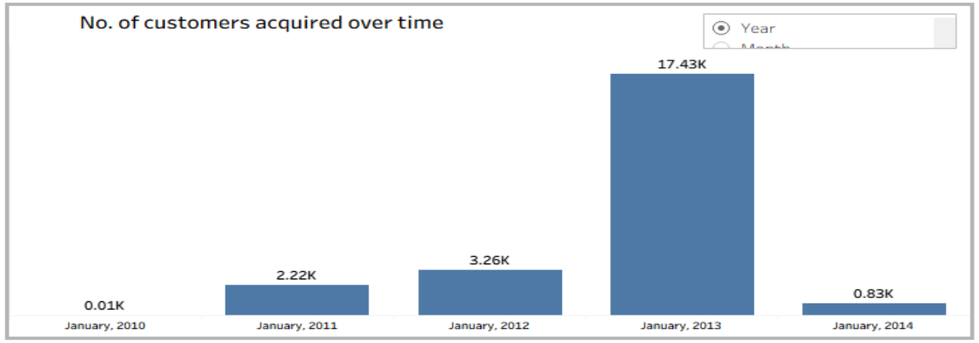
Top 15 Customers Sales Amount Wise



Profession wise Sales Amount



No. of customers acquired over time



Parameter 2
- Sales Amount
- Profit
- Orders

Year of Order Date
- (All)
- 2010
- 2011
- 2012
- 2013
- 2014

English Product Category Name
- (All)
- Accessories
- Bikes
- Clothing
- Components

Data Source | Month Wise sales (7Q). | Year Wise sales (8 Q). | Month wise sales line chart (9 Q) | Quarter wise sales (10 Q). | (11Q) sales amount & Productio... | 12 Q Product Analysis Dash... | 12 Q Customer Analysis Da... | 12 Q Regional Analysis Dash... | 13 Q Final Dashboard New

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Entire View | Show Me

Dashboard | Layout

Device Preview | Device type: Default

Default
Phone
Device Preview

Size
Automatic

Sheets
- Month Wise ...
- Year Wise sale...
- Month wise ...
- Quarter wise ...
- (11Q) sales ...
- Total regions
- Total ...
- Sales Amount
- Total profit
- Total Orders
- Total Products
- Region wise ...
- Region wise ...
- Map region wi...
- Sheet 33
- Region wise ...
- product ...
- Customer ...

Objects
- Horizontal Container
- Vertical Container
- Text
- Extension
- Pulse Metric
- Image
- Blank
- Workflow
- Web Page

Tiled | Floating

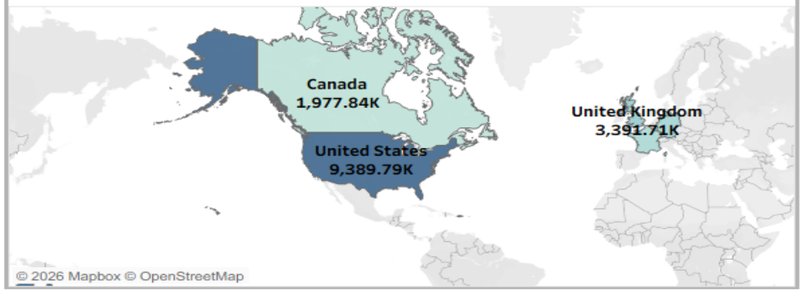
Show dashboard title

Region Analysis Dashboard

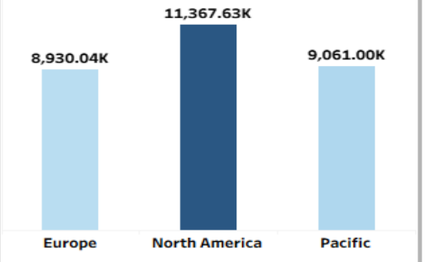
Total Regions	Total Customers	Sales Amount	Total Profit	Total Orders	Total Products
10	18.48K	29.36M	12.08M	60.40K	606

Product Analysis
Customer Analysis

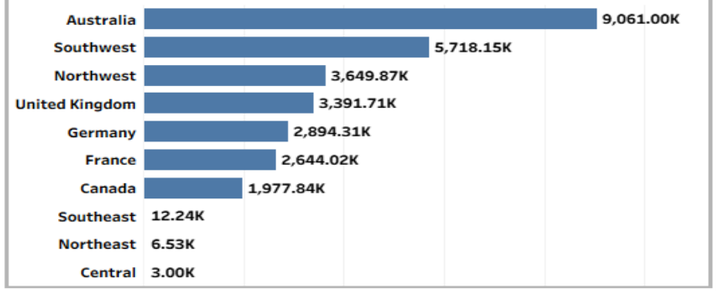
Country wise Sales Amount



Territory Group wise Sales Amount



country wise Sales Amount

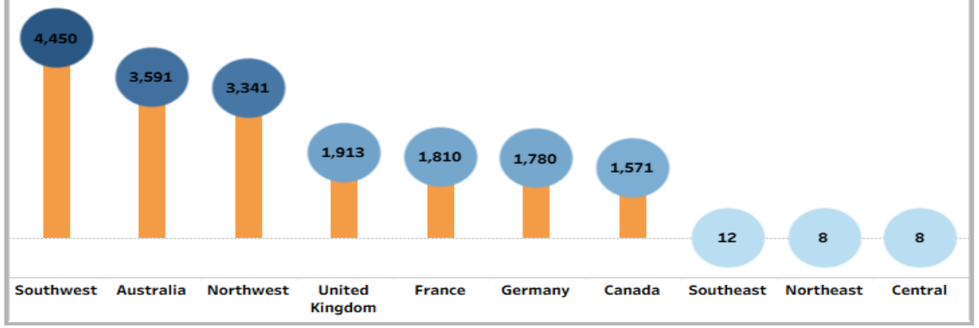


Parameter 2
- Sales Amount
- Profit
- Orders

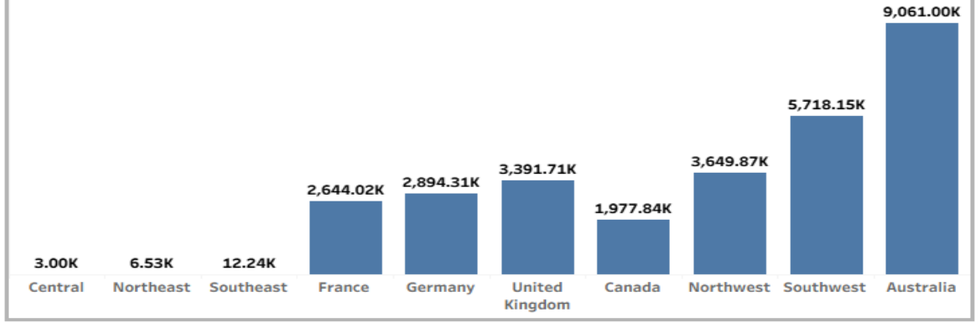
Year of Order Date
- (All)
- 2010
- 2011
- 2012
- 2013
- 2014

English Product Category Name
- (All)
- Accessories
- Bikes
- Clothing
- Components

Region wise Customers



Territory wise Sales Amount



Data Source | Month Wise sales (7Q). | Year Wise sales (8 Q). | Month wise sales line chart (9 Q) | Quarter wise sales (10 Q). | (11Q) sales amount & Productio... | 12 Q Product Analysis Dash... | 12 Q Customer Analysis Das... | 12 Q Regional Analysis Das... | 13 Q Final Dashboard New

1 mark 1 row by 1 column SUM of CNT(Dim_Customer): 18.484

Tableau Public – Tableau Adventure Works project

File Data Worksheet Dashboard Story Analysis Map Format Server Window Help

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Entire View | Show Me

Dashboard | Layout

Device Preview | Device type: Default

Default
Phone
Device Preview

Size
Automatic

Sheets
- Month Wise ...
- Year Wise sale...
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- (11Q) sales ...
- Total regions
- Total ...
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- Total profit
- Total Orders
- Total Products
- Year Wise sale...
- Quarter wise ...
- Month Wise ...
- Day
- (11Q) sales ...
- Profession wis...
- Sheet 33

Objects
- Horizontal Container
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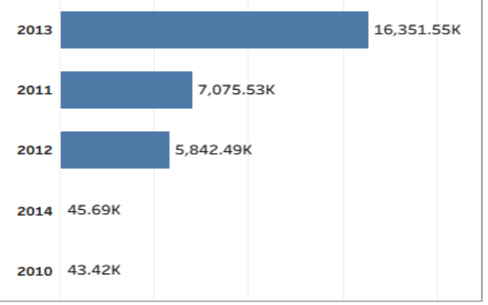
Show dashboard title

Adventure works Analysis by Sales Amount

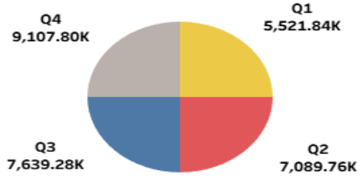
Total Regions	Total Customers	Sales Amount	Total Profit	Total Orders	Total Products
10	18.48K	29.36M	12.08M	60.40K	606

Product Analysis
Customer Analysis
Region Analysis

Year Wise Sales Amount



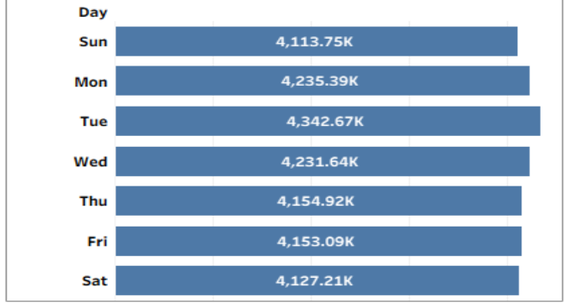
Quarter wise Sales Amount



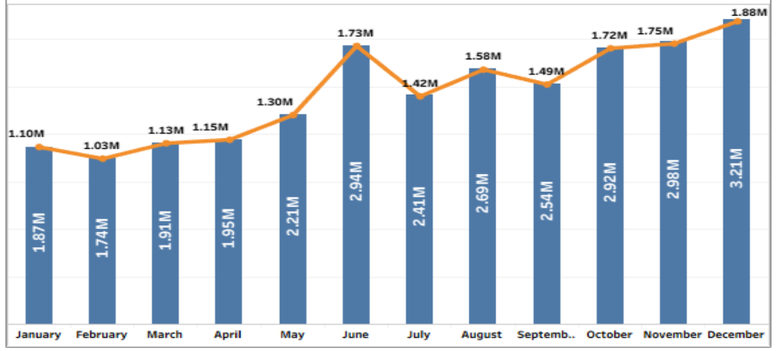
Month Wise Sales Amount

January	1,868.6K
February	1,744.7K
March	1,908.6K
April	1,948.4K
May	2,205.2K
June	2,936.2K
July	2,413.0K
August	2,689.5K
September	2,536.8K
October	2,916.7K
November	2,979.4K
December	3,211.7K

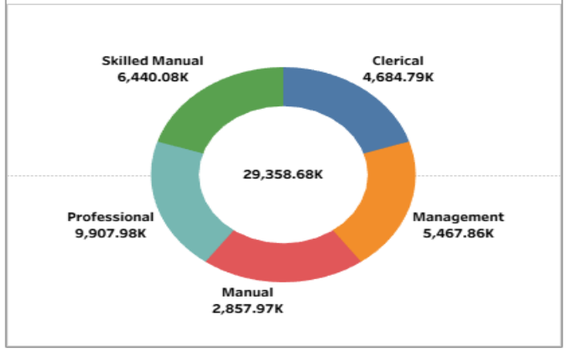
Day wise Sales Amount



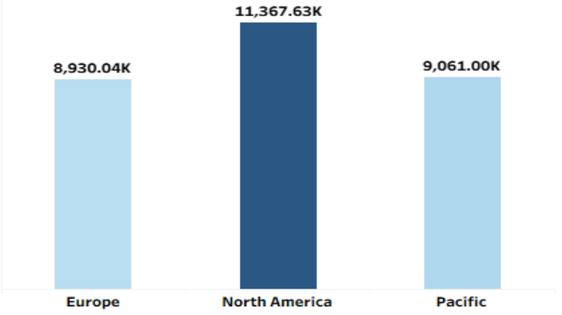
Sales amount (Bar) & Production cost (Line)



Profession wise Sales Amount



Territory Group wise Sales Amount



Parameter 2
- Sales Amount
- Profit
- Orders

English Product Category Name
- (All)
- Accessories
- Bikes
- Clothing
- Components

Year of Order Date
- (All)
- 2010
- 2011
- 2012
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- 2014

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1 mark | 1 row by 1 column | SUM of CNTD(Sales Territory Region): 10

Sql Analysis

- Product Performance Analysis
- Identified Top 5 revenue-generating products
- Found revenue concentration among limited products
- Highlighted high-demand product subcategories
- Customer Behavior Analysis
- Identified Top 3 high-value customers
- Calculated average order value per customer
- Measured total orders per customer
- Time-Based Trend Analysis
- Year-wise sales growth pattern
- Monthly sales trend for 2013
- Identified peak performance year
- Operational Efficiency
- Calculated average shipping time
- Measured delay between order date and ship date
- Evaluated fulfillment performance

• Total Orders: Calculated overall transaction volume using COUNT()

• Maximum Sales Value: Identified highest single order using MAX()

• Top 5 Products: Ranked products based on total sales using GROUP BY sorting

• Category-wise Sales: Used multi-table JOIN to calculate total sales per product category

• Top 3 Customers: Identified high-value customers using average order amount

• Year-wise & Monthly Trends: Analyzed sales growth using YEAR() and MONTH() functions

• Order Processing Efficiency: Measured average shipping time using DATEDIFF()